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Multi-task learning neural framework for categorizing sexism

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ABSTRACT

Sexism, a form of oppression based on one's sex, manifests itself in numerous ways and causes enormous suffering. In view of the growing number of experiences of sexism reported online, automatically classifying these recollections can aid in the battle against sexism by allowing gender studies researchers and government officials involved in policymaking to conduct more effective analyses. This paper investigates the 23-class fine-grained, multi-label classification of accounts (reports) of sexism. Moreover, we propose a knowledge-based cascaded multi-task framework for fine-grained multi-label sexism classification. We leverage several supporting tasks, including homogeneous and heterogeneous auxiliary tasks. Homogeneous tasks are set up without incurring any manual labeling cost and heterogeneous tasks are set up which have a high correlation with the accounts of sexism. Unlabeled accounts of sexism are utilized through unsupervised learning to help construct our multi-task setup. In addition, we incorporate a knowledge module within the framework to infuse external knowledge features into the learning process. Further, we investigate transfer learning that employs weakly labeled accounts of sexism and transfers the learning to the multi-label sexism classification. We also devise objective functions that exploit label correlations in the training data explicitly. Multiple proposed multi-task methods outperform the state-of-the-art multi-label sexism classification across five standard metrics.

1. Introduction

Sexism, as prejudice, stereotyping and discrimination on the basis of sex, is permeated in different open and subtle forms and professional spaces as well. While men and boys are also harmed by sexism, women and girls endure the brunt of sexist mindsets and resultant misconduct. Online forums enabled victims of sexism to publicly express their experiences or witnessed by them by promoting anonymity and bringing distant persons into connection. On the other hand, social media also have content that perpetrates sexism. Thus, the automatic classification of these posts into well-conceived sexism categories can assist to tackle this oppression, as it can better equipping policy makers and gender studies researchers to analyze sexism.

The detection of sexism differs from and can complement the classification of sexism. In a forum where there are posts unrelated to sexism and instances of sexism are present, sexism detection can be used to identify the posts on which to perform sexism classification. While some hate speech classification work (Badjatiya et al., 2017; Waseem and Hovy, 2016) detects sexist hate as a category of hate, but it ignores other forms of discrimination which does not include hate and does not perform sexism classification. We also note the prior work on detecting or classifying personal stories of sexual harassment and/or assault (Chowdhury et al., 2019; Karlekar and Bansal, 2018). In this paper, we focus on classifying an account (report) of sexism involving any set of categories of sexism.

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Table 1
An account of sexism associated with multiple categories.

Account	"I am in social services. my male coworker and I have the same job title. Everyone assumes I am his assistant or an intern. It's not just clients who assume he's my boss, it's other agencies. The same ppl who think nothing calling me honey at work".
Associated categories of sexism	Role stereotyping: False generalizations about some roles being more suitable for women; also applies to similar mistaken notions about men
	Hostile work environment: Sexism suffered at the workplace; also applies when sexism perpetrated by a colleague elsewhere makes working worrisome for the victim
	Sexual harassment (excluding assault): Any sexually objectionable behavior

Most of the existing research on sexism classification (Anzovino et al., 2018; Jafarpour et al., 2018; Jha and Mamidi, 2017) considers at most five categories of sexism. Further, the majority of prior approaches associate only one category of sexism with an instance of sexism. Having mutually exclusive categories of sexism is unreasonable and limiting, as substantiated by Table 1. The example describes an experience of sexism wherein the victim was subjected to three types of sexism. As this is happened in work place it is labeled as *Hostile work environment*, as there is discussion about the roles, it is marked as *Role stereotyping*. The victim was subject to the sexual harassment, as the people calling honey at workplace.

Parikh et al. (2019) explores the multi-label categorization of accounts involving any type(s) of sexism. They provides the largest dataset containing accounts drawn from 'Everyday Sexism Project'.[1] It contains about 13K textual accounts labeled with at least one of 23 categories of sexism formulated with the help of a social scientist. However, they perform sexism classification among 14 categories derived by merging some sets of categories. This prohibits distinguishing within category pairs such as {moral policing, victim blaming} and {motherhood-related discrimination, menstruation-related discrimination}. To overcome this limitation, Abburi et al. (2020a,b) carried out a fine-grained 23-category classification and generated the augmented data through self-training approach. In this paper, we explores the 23-category sexism classification for the multitask classification, where the categories include various fine-grained classes of sexism such as moral policing or role stereotyping along with the widely known categories such as body shaming, rape, and threats. Table 1 defines a subset of the 23 categories; the entire set is listed in Fig. 5.

Given the limited labeled data and large number of categories for our sexism classification, complementary signals for the learning may be beneficial. As far as we know, this paper presents the first multi-task approach for any type of sexism classification. We propose a semi-supervised multi-task multi-label classification approach involving (up to) two broad tasks, namely homogeneous and heterogeneous. For homogeneous auxiliary tasks, we explore Intra-domain specific data for all the three tasks which are set up automatically, requiring no manual labeling effort; the labels and/or samples needed are created through unsupervised learning or acquired via weak labeling. We obtain unlabeled accounts of sexism from 'Everyday Sexism Project' for unsupervised topic proportion distribution estimation and clustering and for sexism detection weak labeling. For heterogeneous auxiliary tasks, we explore the cross-domain data related to the sarcasm detection and emotion classification which has high correlation with the accounts of sexism. Heterogeneous tasks are framed keeping in mind that the accounts of sexism may have the sarcastic and emotion information, so the sarcastic and emotion cues can help the model to perform better.

We develop a knowledge-based neural cascaded multi-task architecture that allows for shared learning across multiple tasks through common layers/weights and a combined loss function. As the model may not understood the domain-specific keywords, we fed the domain specific external knowledge representation into the learning process in order to enhance the performance of our multitask system.

We also develop a weakly labeled transfer learning (two-level training strategy) for multi-label sexism classification since there is a scarcity of labeled data for a few categories in the 23 categories of sexism. In the unlabeled data crawled from the Everyday Sexism Project, a few posts are associated with weakly labeled tags. So we leverage that data by training the model on weakly labeled data as supervised pre-training step and initializing the weights of the desired fine-grained sexism classification model with the first level model to benefit from more data.

In a multi-label setting, the presence of one label may affect the likelihood of the presence of another. While the label co-occurrences in the training data are indirectly accessible to the standard classifier training (since the entire training data is used during the course of an epoch), we propose principled ways of utilizing them explicitly. We formulate loss functions for multi-label classification relating to pair-wise label correlations. We compute pair-wise label co-occurrence statistics from the training data and use them as targets for the proposed objective functions. Through our loss functions, we seek to bring together the actual conditional co-occurrence probabilities and the corresponding model-based probability estimates. In addition to a generic loss function, we present a loss focusing on only non-co-occurring label pairs. We combine a proposed semi-supervised knowledge-based cascaded multi-task neural model with weakly labeled transfer learning and our generic loss function to form a method that outperforms numerous baselines with a clear margin. Our key contributions are summed up below.

- We propose a knowledge-based cascaded multi-task neural framework for the multi-label classification of accounts of sexism that allows for shared learning across multiple tasks through common layers/weights and a combined loss function.

[1] https://everydaysexism.com.

- We explore two types of auxiliary tasks: (a) Homogeneous tasks — in which three appropriate auxiliary tasks are automatically prepared through unsupervised learning and weak labeling and (b) Heterogeneous tasks — in which two tasks are considered to provide external cues from cross-domain data.
- We developed a transfer learning approach where we train the model with weakly labeled data and transfer the knowledge to the desired multi-label sexism classification.
- We propose loss functions aimed at tapping label correlations in the multi-label data explicitly.
- Our best-performing multi-task method, loss function and knowledge transfer learning yield better results than many diverse baselines across five different metrics individually.

The rest of our paper is structured as follows. Section 2 reviews related prior work. Section 3 describes how we formulate the auxiliary tasks, multi-task neural architecture and the weakly labeled transfer learning. Section 4 provides the loss functions that are explored for multi-label sexism classification. Section 5 discusses datasets, experimental results and provides analysis. We conclude with a summary and potential directions for future work in Section 6.

2. Related work

In this section, we describe prior work on the classification of sexism. Although our work involves accounts of sexism, existing work on the classification of misogynous or sexist statements (tweets in which one perpetrates misogyny or sexism) and some distantly related work relating to the classification and identification of hate speech, are also included in this review as some of it applies to our work in the detection of sexist hate to some extent. This section concludes with a discussion of multi-task learning.

We note that sexism is detected by some hate speech classification methods that include sexism as a category of hate (Badjatiya et al., 2017; Waseem and Hovy, 2016; Zhang and Luo, 2018; Davidson et al., 2017) as the sexism classification can be preceded by sexism detection to remove posts unrelated to sexism. Davidson et al. (2017) uses standard machine learning to detect hate speech and abusive language in tweets based on crowd-sourced data. Tweets are classified as racist, sexist, or neither (Waseem and Hovy, 2016) using character n-grams along with extra-linguistic features. On the same dataset, Badjatiya et al. (2017) investigated various deep learning approaches such as fastText, RNN, and CNN to classify the given tweet as sexist, racist, and neither and demonstrated that deep learning models outperformed the basic machine learning models. Zhang and Luo (2018) explored the word embeddings with a combination of CNN, skipped CNN and GRU to classify tweets as sexism, racism, both, and non-hate. Frenda et al. (2019) presents an deep learning approach for detecting the two sides of patriarchal behavior such as sexism and misogyny from tweets. Gao et al. (2017) detects hate speech in a weakly supervised setting where sexist is one of the categories using a two-path bootstrapping approach. Nobata et al. (2016) extracts linguistic, semantic, syntactic, and character n-grams features from the corpus of Yahoo Finance and News comments in order to detect abusive comments and analyze the abusive language over time. Burnap and Williams (2016) use a dictionary, bag-of-words, and text parser to extract typed dependencies inorder to build a data-driven model of cyberhate to identify sexual orientation, disability, and race. Given our emphasis on sexism classification, we do not delve too deeply into previous work on cyberbullying. Van Hee et al. (2015), Agrawal and Awekar (2018) and Zhong et al. (2016).

There is also research being done in multilingual hate speech and sexism detection, in which researches create datasets in various languages. Corazza et al. (2020) investigates different datasets from various languages to examine multilingual hate speech detection. Plaza-Del-Arco et al. (2020) performs hatespeech detection in Spanish tweets for the partially superposed domains of xenophobia and misogyny by comparing various machine learning and deep learning approaches. Rodríguez-Sánchez et al. (2020) create the sexism dataset in Spanish and explored BILSTM and BERT models to detect the sexism in tweets. Chiril et al. (2020b) created a accounts of sexism dataset in French, where the accounts are crawled from the everyday sexism project. They proposed a method for detecting sexism reports/denunciations from actual sexist material that is specifically addressed to a target or defines a target. Chiril et al. (2020a) also created an sexism detection dataset in French language where the posts were crawled from the Twitter and explored several deep learning architectures such as CNN, CNN–LSTM, bi-LSTM with attention, BERT. Guellil et al. (2021) created a sexism dataset in the Arabic language. To validate the corpus, a deep Convolutional Neural Network (CNN), long short-term memory (LSTM) network, and Bi-directional LSTM (Bi-LSTM) networks are explored. In this work, we primarily focus on the accounts of sexism written in English.

Melville et al. (2018) employs topic modeling to map the semantic relationships between topics on sexism data which is obtained from the Everyday Sexism Project. ElSherief et al. (2017) investigates user engagement with posts about gender based violence and their linguistic nuances. A density matrix encoder inspired by quantum mechanics (Yan et al., 2019) is used for the classification of personal stories of sexual harassment. For classifying personal experiences of sexual harassment into one or more of three categories (Karlekar and Bansal, 2018; Chowdhury et al., 2019) explore RNN, CNN, and CNN+RNN models. The deep learning models are explored (Khatua et al., 2018) to classify sexual violence into four fine-grained categories. Sharifirad et al. (2019) explored convolution neural networks for categorization of sexism that includes indirect, sexual, and physical sexism. In Anzovino et al. (2018), tweets identified as misogynist are classified as sexual harassment and threats of violence, discredit, stereotype and objectification, dominance, or derailing using features such as text embedding, n-grams, Part of Speech (POS) tags. Suvarna and Bhalla (2020) identified victim blaming language on Twitter using the transfer learning based classification method. Jafarpour et al. (2018) perform a 4-class categorization of sexist tweets. In Jha and Mamidi (2017), tweets are classified as benevolent, hostile, or non-sexist using biLSTM with attention, SVM, and fastText. While its categorization of sexism pertains to how it is expressed, our research focuses on what an instance of sexism entails, where it occurs, and who perpetrates it.

The majority of the preceding works investigate a dataset with 2 to 5 mutually exclusive categories. Parikh et al. (2019, 2021) explore multi-label categorization of accounts reporting any kind(s) of sexism. They provide the largest dataset for sexism

classification in a multi-label setup, with 23 categories. However, they merged a few categories which have fewer posts, resulting in a total of 14 categories. They experimented with 14 categories with there proposed neural approach which can combine sentence embeddings from pre-trained models like BERT with those generated using biLSTM with attention and CNN. As merging the categories prohibits distinguishing within category pairs, Abburi et al. (2020a, 2021) explores the fine-grained 23 categories of sexism and proposed a semi-supervised data augmentation approaches to augment the original labeled data for the multi-label classification of accounts describing any type(s) of sexism. In this research, we explore the fine-grained 23 categories to perform multi-task learning sexism classification. As far as we know, our work presents the first semi-supervised Multi-Task Learning (MTL) approach involving different auxiliary tasks that are automatically set up without any manual effort for the multi-label classification of accounts describing any type(s) of sexism.

Multi-Task Learning is inspired by human learning activities wherein people often apply the knowledge learned from previous tasks to help learn a new task (Caruana, 1997; Zhang and Yang, 2017). MTL is related to other learning instances in machine learning, including transfer learning (Tan et al., 2018), and multi-label learning (Zhang and Zhou, 2014). MTL is set up similarly to transfer learning, but there is a significant difference. Each data point in multi-label learning is associated with multiple labels, which can be categorical or numerical. If we consider each of the possible labels to be a task, multi-label learning can be thought of as a subset of multi-task learning in which different tasks always share the same data during the training and testing phases. Multi-task learning for multi-label text classification is explored for several problems such as text classification (Zhang et al., 2021; Zhao et al., 2020), emotion recognition (Bendjoudi et al., 2020), and summarization (Elnaggar et al., 2018) etc. It is useful for multiple (related) tasks to be learned jointly so that the knowledge learned in one task can benefit other tasks. There has been considerable interest in applying MTL to a variety of problems including text classification using deep neural networks (DNNs) (Liu et al., 2019; Xu et al., 2019; Guo et al., 2018; Ruder et al., 2019). MTL also increases performance in conjunction with semi-supervised learning (Liu et al., 2007), reinforcement learning (Gupta et al., 2020), and adversarial learning (Liu et al., 2017). MTL provides an effective way of leveraging labeled data from auxiliary tasks for the core task, especially when labeled data available for single-task learning is not large. Recently, the authors of Sawhney et al. (2021) identifies the sexual abuse disclosure narratives in online posts as a joint modeling task that leverages their emotional attributes through multitask learning. They demonstrated that the positive knowledge transfer through context-specific shared representations of a flexible cross-stitched parameter sharing model contributes to establishing the inherent benefit of jointly modeling tasks related to sexual abuse disclosures with emotion classification from the text. In this work, we adopt MTL for fine-grained multi-label sexism classification using several auxiliary tasks including sarcasm detection and emotion classification tasks.

3. Proposed semi-supervised multi-task approach for sexism classification

Our problem statement is to classify an account of sexism (also referred to as a post henceforth) into one or more of 23 categories of sexism. This section introduces a semi-supervised multi-task approach for it. We begin with the auxiliary task setup, architecture details and then provide the weakly labeled transfer learning. Henceforth, we refer to the labeled training set as L and the unlabeled accounts of sexism set as U (w.r.t. categories of sexism).

3.1. Formulating auxiliary tasks

We explore two types of auxiliary tasks, namely homogeneous (intra-domain) and heterogeneous (cross-domain) tasks, that could complement sexism classification in learning terms. In homogeneous tasks, we construct three auxiliary tasks which make use of the unlabeled accounts of sexism obtained from 'Everyday Sexism Project' that substantially outnumber the labeled instances available. The labels for all these tasks and additional samples for one are obtained through unsupervised learning or weak labeling, as depicted in Fig. 1 and detailed below. In heterogeneous tasks, we investigate the two well-defined tasks, which have a high correlation with the accounts of sexism to provide more cues for the sexism classification.

Homogeneous Tasks:

3.1.1. Estimating the topic proportion distribution

The *Topic-p* task is the prediction of topic proportion distributions, signifying the degrees to which a given post relates to a fixed number of topics. For estimating these distributions (which act as target labels in our subsequent multi-task learning) and topics, we employ lda2vec (Moody, 2016). It produces post-to-topic proportions by mixing word2vec's skip-gram architecture with Dirichlet-optimized sparse topic mixtures. We fix the number of topics to 10. While we train lda2vec on $L \cup U$, we use only the topic proportion distributions associated with L for multi-task learning. Experiments that substitute them with their U-based counterparts underperform.

3.1.2. Predicting the cluster label

Another auxiliary task that we explore is the prediction of the cluster to which a given post belongs (*Cl-pred*). We perform k-means clustering on vector representations of the posts in $L \cup U$ and thereby augment L with cluster labels for setting up this k-class classification. The post representations are created using a BERT (Devlin et al., 2018) model tuned using unlabeled accounts of sexism (henceforth called *BERT-t*). The number of clusters k is a hyper-parameter that we tune.

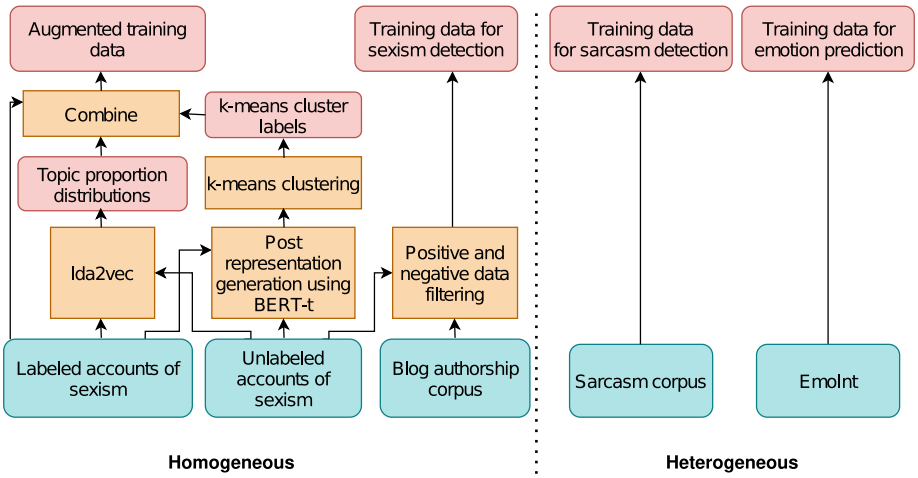

Fig. 1. Unsupervised data creation for multi-task sexism classification.

3.1.3. Detecting an account of sexism

Identifying whether a given post is an account of sexism is adopted as a third task (*S-det*). We obtain the (weakly labeled) negative data for this task from the Blog Authorship Corpus (Schler et al., 2006) through two types of filtering. To help select accounts as opposed to commentary, the presence of a few keywords and one past tense POS tag is mandated. Moreover, we exclude posts that exceed the word and sentence count maximums of L. This filtering is also used while randomly selecting posts from U as the positive data. L itself can also be (additionally) used for this purpose.

We also experimented with a way of generating the weakly labeled negative data for the task of detecting an account of sexism that involved additional filtering. We sought to only include accounts of a form of oppression other than sexism in the negative data for this experiment. Using inputs from a social scientist and the lists of ethnic[2] and LGBT-related[3] slurs from Wikipedia, we compiled a list of terms related to forms of oppression other than sexism. The posts from the Blog Authorship Corpus that do not contain any word from this list are not considered to be included in the negative data for sexism detection. The motivation behind this filtering is to see if the negative data obtained through it, comprising entirely of accounts of oppression but not of sex-based oppression, constitutes a harder negative set and improves the performance. As this did not performed well, we did not report the results.

Heterogeneous Tasks:

3.1.4. Detecting sarcasm

Detecting whether a given post is sarcastic or not is considered as one of the heterogeneous auxiliary task (*sarc-det*). We obtain data from the Sarcasm corpus (Oraby et al., 2017), which contains both positive and negative data. As the sarcastic text is present in the accounts of sexism, this task is designed to help predict sexism accounts more accurately by transferring sarcastic cues to the sexism classification model.

3.1.5. Predicting the emotion

Another heterogeneous auxiliary task that we explore is predicting the emotion to which a given post belongs (E-pred). For this, we obtain data from the EmoInt (Mohammad et al., 2018), in which each post is associated with at least one of the 11 categories. This task is motivated by the fact that the emotional attributes have a high correlation with the narratives describing the instances of sexism. As a result, these emotional cues may aid in predicting the accounts of sexism more accurately.

3.2. Proposed architecture

Our knowledge-based cascaded multi-task neural architecture is depicted in Fig. 2. We divided our architecture into three parts, namely Block A, knowledge module and cascading module.

[2] https://en.wikipedia.org/wiki/List_of_ethnic_slurs.
[3] https://en.wikipedia.org/wiki/List_of_LGBT-related_slurs.

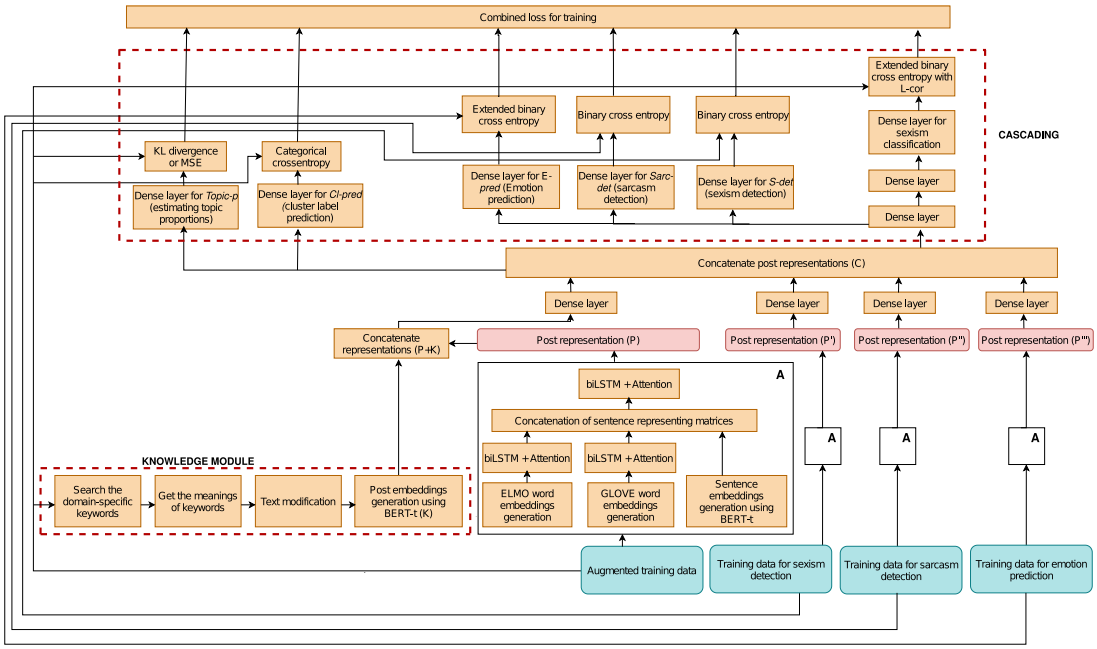

Fig. 2. Proposed knowledge-based cascaded multi-task neural classification.

3.2.1. Block A

If the list of chosen tasks consists of sexism detection and the heterogeneous auxiliary tasks such as sarcasm detection and emotion prediction; the training input is a batch of tuples (samples), each of which consists of a post from L and $|L|$ posts randomly picked from the sexism detection training data and $|L|$ posts from the heterogeneous tasks training data each. Each of these posts in a sample is processed hierarchically to produce the corresponding post representations (P, P′, P″, P‴) using the same set of layers/weights (block A in Fig. 2). First, the words of each sentence (in each post) are embedded separately using ELMo (Peters et al., 2018) and GloVe (Pennington et al., 2014). For each of the two resulting three-dimensional tensors, the word vector matrix for each sentence is passed through biLSTM linked with an attention scheme (Yang et al., 2016). Overall, this yields two sentence representing matrices, providing two vector representations for each sentence. This is complemented by another embedding generated for each sentence using *BERT-t*. Next, the concatenation of the three sentence representing matrices is fed to biLSTM + attention to create the post representation.

3.2.2. Knowledge module

To enhance the performance of our multi-task system, we incorporate domain-specific external knowledge representation (K) to the existing architecture that introduces more knowledge information to the learning process. To begin, we compiled a list of domain-specific keywords under the supervision of a social scientist. List of domain-specific keywords are present in the link.[4] In each post, we search for the domain-specific keywords that are present in the list and get their corresponding meanings using PyDictionary[5] based on the POS tag of the word (as the same word has a different meaning based on the context, we used POS tag). We modify the text of each post by replacing the domain-specific keywords with their meanings and leaving out words that are not in the list of domain-specific keywords as it is. For the modified text, a knowledge representation (K) is produce using *BERT-t*. The post representation (P), which is the output of the augmented training data (accounts of sexism) and knowledge representation (k) are concatenated to produce knowledgeable post representation (P+K).

Further, the knowledgeable post representation (P+4K) and other post representations (P′, P″, P‴) are fed into the dense layer (relu) separately. Each of the four resulting representations are concatenated to create the final post representation (C).

3.2.3. Cascading module

Instead of learning a single representation, we generate the output of different tasks at different levels (where the dense layer is introduce at each level) in a cascading setup. We build a multi-level cascaded multi-task system on top of the final post representation (C), where at each level, the output is fed to the auxiliary tasks or primary task and dense layer (relu). At the first level, the final vector representation (C) is passed to (1) a fully connected layer which is custom-made for each of *Topic-p*, *Cl-pred* and to the (2)

[4] https://github.com/Harikavuppala1a/Semi-supervised_knowledge-based_multi-task-learning/blob/main/data/domain_specific_keywords.txt.

[5] https://github.com/geekpradd/PyDictionary.

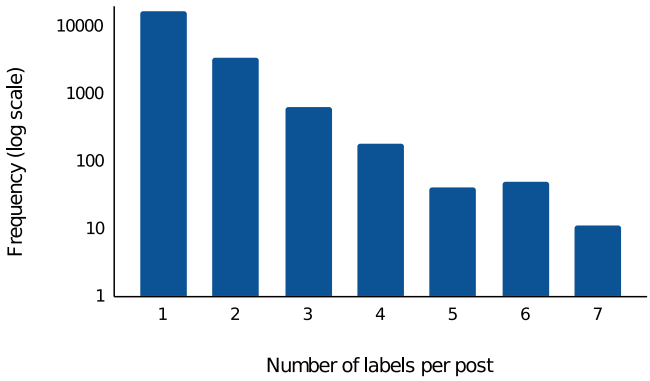

Fig. 3. Frequency distribution of number of labels per post.

dense layer. At the second level, the output of dense layer is fed to a dense layer design for *E-pred*, *Sarc-det*, *S-det*, and another dense layer. At the third level, the dense layer output is fed into a fully connected layer design specifically for sexism classification. As the *S-det* auxiliary task is closest to the sexism classification (primary) task and the heterogeneous tasks like *E-pred* and *Sarc-det* provide sarcastic and emotional cues to the primary task, we keep them at the second level, which is close to the final layer rather than at the first level.

For sexism classification, *S-det*, *Sarc-det*, and *E-pred*, the sigmoid activation is used; for the other two tasks, we use softmax. For the loss for *Topic-p*, we explore KL divergence (as lda2vec outputs a probability distribution) but find mean squared error (MSE) more effective. For *Cl-pred*, we use categorical cross entropy loss. For *S-det* and *Sarc-det*, we explore binary cross entropy losses and for *E-pred*, we use extended binary cross entropy loss (Parikh et al., 2019). For sexism classification, we employ the extended binary cross entropy label correlation loss (EBCEL-cor) addressing the multi-label aspect as well as explicitly consider label correlations that exist in a multi-label data, discussed in Section 4. We incorporate class imbalance neutralizing weights into the loss functions where needed. The final loss we train is a weighted combination of these losses, where the weights are hyper-parameters.

We also explore single level cascading setup, where the post representations corresponding to *S-det*, *Sarc-det* and *E-pred* are fed to their corresponding custom-made fully connected layers. The knowledgeable post representation (P+K) is fed to the dense layer (relu) and to the other two tasks such as *Topic-p*, *Cl-pred*. The output of the dense layer is then fed to a dense layer design for sexism classification. It is evident from the results that the multi-level cascading produce better results than the single-level variant; hence we consider the multi-level cascaded multi-task set up to build our proposed approach.

We experiment with variants of this architecture corresponding to some subsets of all the auxiliary tasks that are described. We also attempt transfer learning with this architecture wherein a variant of the proposed model comprising only one auxiliary task's dense layer is used to pre-train the weights of block A used in another variant meant for the core sexism classification task. Since this does not perform well, it is omitted.

3.3. Weakly labeled transfer learning

We devise a Weakly Labeled Transfer Learning (WLTL) approach (two-level training strategy) for multi-label sexism classification wherein we train a model using weakly labeled data and then transfer that knowledge to the main sexism classification task. The motivation behind this transfer learning is the limited labeled data for some categories for our 23-class fine-grained categorization scheme, and also a large amount of freely available weakly labeled accounts of sexism data. As the similarity scores for posts associated with the weakly labeled tags and labeled categories are high (greater than 0.7) for most of the categories, we use the training on weakly labeled tags as a supervised pre-training step. The intuition is that, as the weakly labeled data is more than labeled data, this could lead to better weights, and initializing the final model with those weights as opposed to random weights could yield a superior fine-grained classifier. The architectures of the models at both levels are the same except for the final dense layer. At each level, the dense layer of each model has the same number of output units as the category set size.

When the unlabeled data is crawled from the Everyday sexism project, we observed that a few posts are associated with one or more tags (categories) which are provided by the survivors or observers when they share their experiences. Out of 90,000 unlabeled accounts of sexism, 21,000 posts have the tags associated with it (after removing the labeled data). Among these tags, we found 2479 unique tags where most of the tags have fewer posts associated with them. So we created the "Other" category and merged all the categories which have fewer posts to the Other category, resulting in 7 categories: 'Home', 'School', 'Public space', 'Workplace', 'University', 'Media', and 'Other'. Fig. 3 shows the frequency distribution of the number of labels per account in the weakly labeled dataset, demonstrating the multi-label nature of instances of sexism. Table 2 lists 4-grams from the textual accounts of sexism associated with each weakly labeled category of sexism to show what are the most frequent phrases that each category have.

To perform the knowledge transfer, we create a two-level approach. Fig. 4 specifies the two-level hierarchy of labeled and weakly labeled categories of sexism. At the first level, we train our proposed multi-task model on the weakly labeled data with the dense

Table 2
Linguistic analysis of the weakly labeled dataset.

ID	Weakly labeled tags	4-grams
1	Home	'I was walking home', 'to leave me alone', 'hand up my skirt'
2	School	'in front of home', 'boys are trying to', 'boys in my class'
3	Public space	'walking down the street', 'a group of strangers', 'his hand on my'
4	Workplace	'harassment in the workplace', 'the only woman in', 'to a male colleague'
5	University	'I was at university', 'sat next to me', 'tried to kiss me'
6	Media	'a picture of a', 'was watching an episode', 'on the front page'
7	Other	'made me feel really', 'asked me if I', 'to put up with'

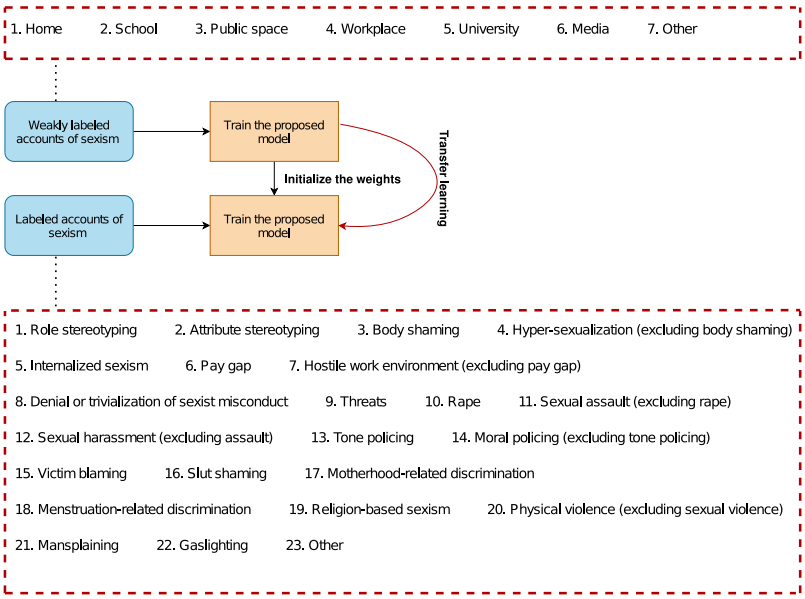

Fig. 4. Weakly labeled knowledge transfer learning for multi-label sexism classification.

layer, with seven output units (as the category set size). At the second level, we train the 23-class fine-grained multi-task sexism categorization model on the labeled accounts of sexism, where we initialize the weights of this model with the previous level model except for the last layer.

4. Label co-occurrence based loss functions for multi-label classification

In this section, we propose loss functions which explicitly use label correlations existing in the multi-label data. In order to take advantage of label correlations this way, we compute statistics pertaining to pair-wise label co-occurrences from the training set L and supply them as targets to the loss functions. First, we compute the symmetric label co-occurrence (count) matrix M. We then estimate the probability of label l_j occurring given the occurrence of label l_i, $P(j|i)$, as $M(i,j)/f(i)$, where $f(i)$ denotes the frequency of label l_i. Next, we compute the set of label pairs correlated to a specified degree; $S = \{i, j \mid P(j|i) \leq t,\ i \neq j,\ 1 \leq i, j \leq q\}$, where the threshold t is a hyper-parameter. The proposed loss function *L-cor* is then given as follows.

$$L\text{-}cor = \frac{1}{|S|} \sum_{(i,j)\in S} \left| \frac{\sum_{k=1}^{n} \hat{p}_{ki}\hat{p}_{kj}}{\sum_{k=1}^{n} \hat{p}_{ki}} - P(j|i) \right| \tag{1}$$

Here, n and q are the numbers of posts and labels respectively. $\hat{p}_{ki}$ is the estimated probability of label l_i applying to post x_k. *L-cor* is aimed at minimizing the L1 norms of the differences between the actual conditional co-occurrence probabilities $P(j|i)$ and their estimated model-based counterparts $\hat{P}(j|i)$ for the label pairs in S. The $\hat{P}(j|i)$ estimates are derived from the probabilities $\hat{p}_{ki}$ output by the model. To include all non-diagonal labels pairs in the loss computation, t is set to 1. Conversely, to impose penalties for only non-co-occurring (non-diagonal) label pairs, we set t to 0 and can remove $\|$ from Eq. (1) (as $\forall (i,j) \in S,\ P(j|i) = 0$ when $t = 0$).

We also devise another loss function targeting non-co-occurring label pairs. We first compute the set of uncorrelated pairs (as per the training data); $S_u = \{i, j \mid M(i,j) = 0,\ i < j,\ 1 \leq i, j \leq q\}$. Using that, our loss function *L-unc* is computed using the following expression.

$$L - unc = \frac{1}{|S_u|} \sum_{(i,j)\in S_u} \frac{(\sum_{k=1}^{n} \hat{p}_{ki}\hat{p}_{kj})(\sum_{k=1}^{n} \hat{p}_{ki} + \sum_{k=1}^{n} \hat{p}_{kj})}{(\sum_{k=1}^{n} \hat{p}_{ki})(\sum_{k=1}^{n} \hat{p}_{kj})} \tag{2}$$

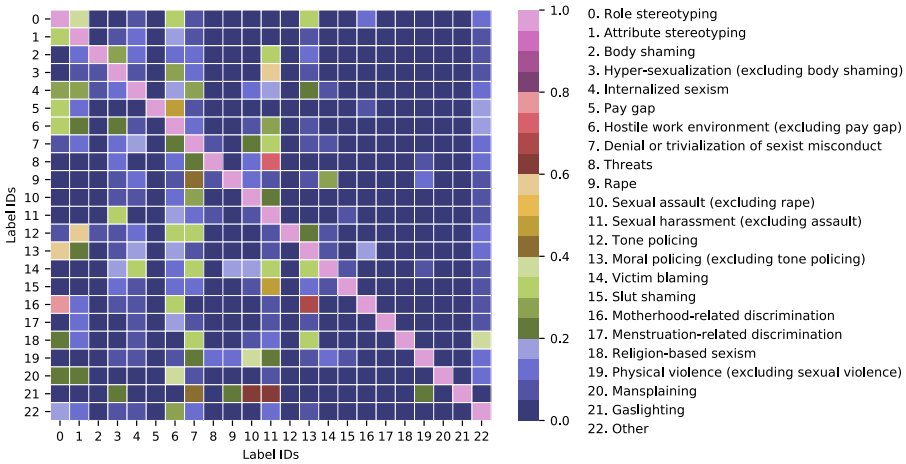

Fig. 5. Pairwise conditional label co-occurrence matrix for the training data.

In effect, for each $(i, j) \in S_u$, we sum the model-based conditional co-occurrence probabilities $\hat{P}(j|i)$ and $\hat{P}(i|j)$ here. Since S_u comprises only non-co-occurring label pairs, these co-occurrence probability estimates should be low for the optimal model weights. Hence, we minimize their sum.

We also explore two variants of the *L-cor* and *L-unc* losses. Replacing the L1 norm with the L2 norm in Eq. (1) constitutes one of them. The other involves replacing the sum of the model-based conditional co-occurrence probabilities in Eq. (2) with a model-based co-occurrence score. Both underperform *L-cor* and *L-unc*.

Each proposed label co-occurrence based loss is weighed by a hyper-parameter (β) and added to extended binary cross entropy (EBCE) loss (Parikh et al., 2019). In the equation *L-cor* can be replaced with *L-unc*.

$$EBCEL-cor = -\frac{1}{n}\sum_{i=1}^{n}\frac{1}{L}\sum_{j=1}^{L} w_{jy_{ij}}\Big\{y_{ij}\log(\hat{p}_{ij}^{\sigma}) + (1-y_{ij})\log(1-\hat{p}_{ij}^{\sigma})\Big\} + \beta * L-cor \tag{3}$$

Here, n and L denote the number of samples (posts) and the number of classes respectively. y_{ij} is 1 if label l_j applies to post x_i and 0 otherwise. $\hat{p}_{ij}^{\sigma}$ is the estimated probability of label l_j being applicable to post x_i computed using a sigmoid activation atop the fully connected layer with L output units. α is a hyper-parameter. The weights for correcting class imbalance w_{jv} are computed as follows.

$$w_{jv} = \frac{n}{2|\{x_i \mid y_{ij} = v, 1 \le i \le n\}|} \tag{4}$$

We use some of the combined loss functions in conjunction with the existing state-of-the-art model as well as our multi-task model.

5. Experiments

This section provides the experimental evaluation of the proposed multi-task sexism classification methods against a number of baseline methods and presents analysis. Our code is available on GitHub[6]

5.1. Datasets

5.1.1. Labeled data

We use the dataset contributed by Parikh et al. (2019) comprising 13,023 accounts of sexism, each labeled with at least one of 23 categories of sexism. The diverse categories of sexism, derived in consultation with a social scientist by considering campaigns and gender-related discourse (Eccles et al., 1990; Dutta and Sircar, 2013; Menon, 2012; Mead, 1963) as well as possible impact on public policy. The categories range from menstruation-related discrimination and body shaming to role stereotyping and victim blaming. Fig. 5 lists the 23 categories and shows the pairwise conditional label co-occurrence matrix for the training data L. We caution that coverage/co-occurrence statistics related to sexism shown in this work do not represent their real-world counterparts.

[6] https://github.com/Harikavuppala1a/Semi-supervised_knowledge-based_multi-task-learning.

5.1.2. Unlabeled data

We obtain 90,000 unlabeled instances of sexism from 'Everyday Sexism Project', which has several hundred thousand accounts of sexism from survivors and observers. First, we remove the instances corresponding to all the labeled (training, validation, and testing) posts from them. Then, we shortlist 70,000 shortest instances containing a minimum of 7 words each to form U. Short posts are preferred to maximize the resemblance to the labeled data.

5.1.3. Blog Authorship Corpus

Additionally, we use Blog Authorship Corpus (Schler et al., 2006) to obtain weakly labeled negative data for the sexism detection auxiliary task; the keywords mandated therein to help select accounts as opposed to commentary are 'i','we', and 'he'. The negative and positive sets each comprise 20,000 posts.

5.1.4. Sarcasm Corpus

The Sarcasm Corpus V2 is a subset of the Internet Argument Corpus (IAC) (Walker et al., 2012) which is a corpus for research in political debate on internet forums. This Sarcasm Corpus V2 dataset consists of 6520 posts that have been labeled as either sarcastic or not-sarcastic.

5.1.5. Emotion Corpus

This dataset has been taken from SemEval-2018 Task-1 (Mohammad et al., 2018) that covers the emotion-specific labels representing the mental state of the authors of the tweets. It consists of 10,986 tweets divided into 11 emotion labels (anger, anticipation, disgust, love, fear, joy, sadness, optimism, pessimism, trust, and surprise), each being a binary label indicating the presence of a specific emotion. Each tweet in the dataset is associated with at least one of the 11 categories, which are set up in a multi-label manner.

5.2. Evaluation metrics

Multi-label classification, wherein classes can co-occur, is evaluated differently from the single-label case. We adopt a number of established metrics, namely instance-based F1 referred to as F_{ins}, instance-based accuracy Acc, F1 macro F_{mac}, F1 micro F_{mic}, and Subset Accuracy SA (Zhang and Zhou, 2014; Parikh et al., 2019).

5.3. Baselines

All deep learning architectures below end with a dense layer with the sigmoid activation and are trained using the extended binary cross entropy loss.

- **Random**:
  For each test sample in training data, labels are selected randomly based on their normalized frequencies.
- **Traditional Machine Learning (TML)**:
  We report the performance using Random Forests (RF), Logistic Regression (LR), and Support Vector Machine (SVM) each applied on two feature sets, such as TF-IDF on word unigrams and bigrams (termed as a Word-ngrams) and the average of the ELMo vectors for a post's words. This gives rise to six combinations: word-ngrams with RF (word-ngrams-RF), word-ngrams with LR (word-ngrams-LR), word-ngrams with SVM (word-ngrams-SVM), ELMO with RF (ELMO-RF), ELMO with LR (ELMO-LR), and ELMO with SVM (ELMO-SVM).
- **Deep Learning (DL)**:
  biLSTM and biLSTM-Attention: The word embeddings for a post are passed through a bidirectional LSTM for biLSTM baseline where as the Output of biLSTM is passed to the attention scheme for biLSTM-Attention baseline (Yang et al., 2016).
  Hierarchical-biLSTM-Attention: This architecture is similar to Yang et al. (2016) where the word embeddings are first passed to biLSTM with attention to generate the representation for each sentence. These sentence embeddings are then fed into the another instance of biLSTM with attention.
  BERT-biLSTM-Attention and USE-biLSTM-Attention: The sentence representations are generated using BERT via bert-as-service (Xiao, 2018) and fed to a biLSTM with attention and the same way is followed for USE (Cer et al., 2018).
  CNN-Kim: In this method similar to Kim (2014), the convolutional and max-over-time pooling layers are applied to the word vectors for each post.
  CNN-biLSTM-Attention: Each sentence's word embeddings in each post are fed through convolutional and max-over-time pooling layers. The resultant representations are then passed through a biLSTM with attention similar to Wang et al. (2016).
  C-biLSTM: This is a variant of the C-LSTM architecture (Zhou et al., 2015) and slightly related to a method used in Karlekar and Bansal (2018). After applying convolution on a word vectors, the feature maps are stacked along the filter dimension to generate a series of window vectors, which are fed to biLSTM.

Table 3
Results for machine learning baselines.

Approach	F_{ins}	F_{mac}	F_{mic}	Acc	SA
Word-ngrams-RF	0.538	0.246	0.482	0.444	0.272
Word-ngrams-LR	0.544	0.188	0.492	0.454	**0.287**
Word-ngrams-SVM	0.453	0.227	0.413	0.331	0.107
ELMO-RF	0.374	0.100	0.330	0.307	0.185
ELMO-LR	**0.576**	**0.261**	**0.535**	**0.475**	0.279
ELMO-SVM	0.546	**0.261**	0.501	0.431	0.206

- **Semi-supervised**:
  BERT-t-biLSTM-Attention: This is the same as BERT-biLSTM-Attention except that the BERT model used is fine-tuned on the unlabeled instances of sexism (Parikh et al., 2019).
  Parikh et al. (2019): The state-of-the-art model concatenates sentence representations obtained using a *BERT-t* with those created from ELMo and GloVe embeddings separately using biLSTM with attention. The combined sentence embeddings are passed through biLSTM with attention and explore Extended Binary Cross Entropy (EBCE) loss.
  Self-training: We adapt self training (Yarowsky, 1995) for multi-label classification. Iteratively, we train (Parikh et al., 2019), use it to pseudo-label unlabeled samples, and augment the training set with those for which the mean of the class-wise model-given probabilities exceeds a threshold.

5.4. Results

The pre-processing steps for all deep learning methods include deleting non-alpha-numeric characters and extra spaces, zero-padding input tensors as appropriate, and lower-casing. While splitting the post into sentences, any sentence with more than 35 words is divided into multiple sentences to ensure that the maximum sentence length of 35 words is met.

We set aside 15% from original labeled data for validation and testing each. Using experiments on a validation set, for each method, we choose the values of hyper-parameters: the LSTM dimension, the attention dimension, the number CNN filters (for baselines), β, loss weights. The validation set was merged into the training set during the testing phase. The hyper-parameters for all the proposed models and baselines are provided in our Github repository.[7]

For all deep learning methods, the number of epochs is set to 10, and the learning rate was set to 0.001. We employ 0.25 dropouts after each input and before the final fully connected layer. We choose Adam as the optimizer while training each model. A batch size of 64 is used. These fixed parameters were constant throughout all the approaches. For each deep learning method, the mean of the results obtained over three runs is given for each metric.

For TML baselines, we show the results (see Table 3) produced using three traditional ML methods: RF, LR, and SVM across two different feature sets: word n-grams, and ELMo average. For LR and SVM, we apply class imbalance correction. Among these combinations, ELMo word vectors with logistic regression yield the best results.

Table 4 provides the results produced by various proposed multi-task methods, objective functions, and several baselines. As expected, the random baseline performs extremely poorly, reflecting the challenging nature of the fine-grained multi-label classification. For the deep learning baselines, we explored both ELMO and glove word embeddings and found that ELMo performance is better than GloVe across multiple baselines and hence reported the results with ELMo word embeddings. We report the DL and semi-supervised baseline results with the EBCE loss. We note that the results are reported for only some of the instances that can arise from the proposed configurable neural framework. The best deep learning baseline is Hierarchical-biLSTM-Attention, and it outperforms its traditional ML counterpart. Overall, the semi-supervised method (Parikh et al., 2019) is confirmed as the best baseline. The BERT model tuned on domain-specific unlabeled instances of sexism named tBERT works better than the vanilla BERT counterpart and USE. Augmented data generated by the self-training approach using (Parikh et al., 2019) as the base classifier performed poorly compared to without augmenting the data.

Most proposed methods outperform all baselines across all metrics. The maximum performance improvement is observed for subset accuracy (SA), which is the most stringent metric. The results for the proposed objective functions capitalizing on label correlations are reported with Parikh et al. (2019) architecture. For all the multi-task methods, results are shown with knowledge module and cascading setup. Out of all the variants, *L-cor* performs best for most of the metrics as it seeks to minimize the L1 norms of the differences between the actual and model-based conditional co-occurrence probabilities. The proposed multitask methods results are shown with *EBCEL-cor* objective function as *L-cor* performs best out of all the variants. Our best performing multi-task method involves all the homogeneous and heterogeneous auxiliary tasks such as *S-det*, *Sarc-det*, *E-pred*, *Topic-p*, *Cl-pred* (along with the primary sexism classification task). From the results, it is also observed that the heterogeneous tasks such as *Sarc-det* and *E-pred* complementary signals (cues from cross-domain data) helped for the better classification. On the best multitask approach, we also show the result with weakly labeled transfer learning (WLTL), which turns out to be the best-proposed method involving all the homogeneous and heterogeneous auxiliary tasks with *EBCEL-cor*.

[7] https://github.com/Harikavuppala1a/Semi-supervised_knowledge-based_multi-task-learning

Table 4
Results for the proposed methods as well as the deep learning (DL), and semi-supervised baselines.

	Approach	F_{ins}	F_{mac}	F_{mic}	Acc	SA
DL baselines	Random	0.035	0.090	0.192	0.022	0.003
	biLSTM	0.627	0.451	0.577	0.472	0.147
	biLSTM-Attention	0.648	0.445	0.597	0.499	0.176
	Hierarchical-biLSTM-Attention	0.664	0.485	0.616	0.516	0.191
	BERT-biLSTM-Attention	0.591	0.397	0.546	0.431	0.089
	USE-biLSTM-Attention	0.566	0.398	0.525	0.402	0.061
	CNN-Kim	0.658	0.481	0.617	0.513	0.195
	CNN-biLSTM-Attention	0.421	0.284	0.387	0.278	0.035
	C-biLSTM	0.485	0.316	0.446	0.328	0.038
Semi-supervised baselines	*BERT-t*-biLSTM-Attention	0.632	0.435	0.580	0.472	0.127
	Self-training	0.704	0.513	0.654	0.557	0.220
	Parikh et al. (2019)	0.714	0.546	0.665	0.572	0.242
Proposed objective functions with Parikh et al. (2019)	*L-unc*	0.716	0.546	0.668	0.575	0.242
	L-cor with $t = 0$	0.711	0.550	0.663	0.570	0.247
	L-cor	0.723	0.550	0.672	0.584	0.264
	L-cor with $t = 1$	0.718	0.559	0.670	0.577	0.245
Proposed multi-task methods with *EBCEL-cor*	**Auxiliary tasks**					
	Topic-p	0.721	0.568	0.674	0.585	0.260
	Cl-pred	0.724	0.571	0.679	0.586	0.267
	Topic-p, Cl-pred	0.725	0.580	0.683	0.591	0.272
	S-det	0.729	0.570	0.681	0.593	0.276
	S-det, Topic-p	0.734	0.581	0.687	0.600	0.290
	S-det, Cl-pred	0.728	0.566	0.688	0.597	0.288
	S-det, Topic-p, Cl-pred	0.726	0.572	0.681	0.591	0.276
	Sarc-det	0.723	0.568	0.675	0.584	0.256
	E-pred	0.724	0.566	0.676	0.587	0.269
	S-det, Sarc-det	0.731	0.576	0.685	0.596	0.283
	Sarc-det, E-pred	0.734	0.579	0.690	0.603	0.297
	S-det, E-pred	0.733	0.579	0.686	0.599	0.285
	S-det, Sarc-det, E-pred	0.732	0.589	0.686	0.597	0.292
	S-det, Sarc-det, E-pred, Topic-p, Cl-pred	0.734	0.592	0.691	0.603	0.297
Best multi-task method with WLTL	*S-det, Sarc-det, E-pred, Topic-p, Cl-pred with WLTL and EBCEL-cor*	**0.740**	**0.595**	**0.697**	**0.612**	**0.309**

Table 5
Results on the augmented data generated by Abburi et al. (2020a).

Approach	F_{ins}	F_{mac}	F_{mic}	Acc	SA
Abburi et al. (2020a)	0.738	0.554	0.691	0.610	0.316
Our best proposed method	0.754	0.587	0.709	0.635	0.338

We also show how well our proposed approach performs on the augmented data generated by the Abburi et al. (2020a) in Table 5. From the results, it is observed that our best proposed multitask method (*S-det, Sarc-det, E-pred, Topic-p, Cl-pred with WLKT and EBCEL-cor*) performs the best on the augmented data as well.

We analyze the performance of the different combinations of auxiliary tasks with and without knowledge and cascading modules. Fig. 6 depicts the F_{mac} and SA for five combinations of auxiliary tasks. For the without knowledge and cascading module setup, in Fig. 2 after generating the post representations (P, P′, P″, P‴), the post representations are directly passed to a fully connected layer and then to the loss functions which are custom-made for each of the tasks. The final loss is trained with the weighted combination of these losses, where the weights are hyper-parameters. The figure demonstrates the relative efficacy of the knowledge and cascading method across different auxiliary tasks.

Fig. 7 compares the class-wise performance of our best-performing model with that of the best baseline (Parikh et al., 2019). We report each label's coverage, i.e. the proportion (%) of samples belonging to that label in the training data, to bring out the imbalanced distribution. For a majority of the classes, the proposed method outperforms the baseline. We note a significant improvement over the baseline for several low-coverage classes.

Fig. 8 shows the improvement in F_{mac} observed with increasing training data. Our best method produces greater improvement than the best baseline throughout and also yields a better result for the lowest training data % (i.e. 50). We analyze the impact of the multi-label nature of the problem on the performance in Table 6. The performance for one run of our best method across different numbers of labels per post (1 to 6) is compared against the counterpart in Parikh et al. (2019). We observe improved performance with the proposed method across all metrics for a majority of the cases.

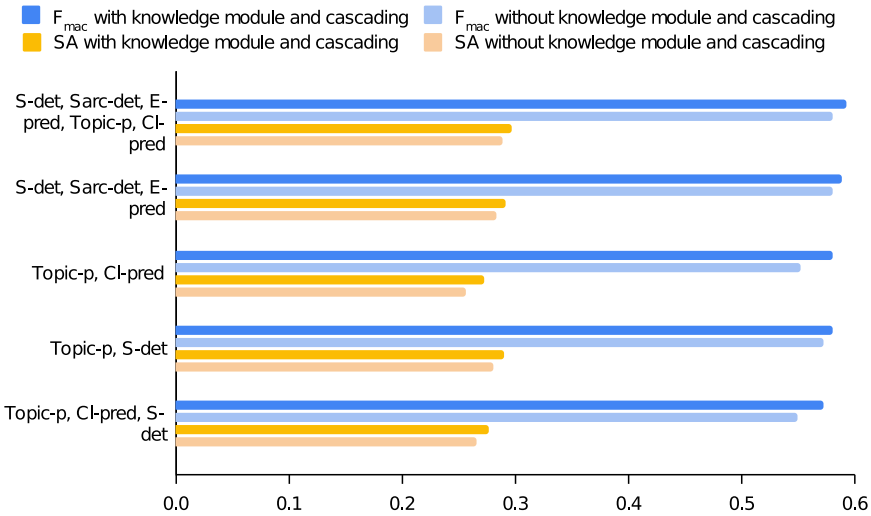

Fig. 6. Performance of different combinations of auxiliary task with and without knowledge and cascading modules.

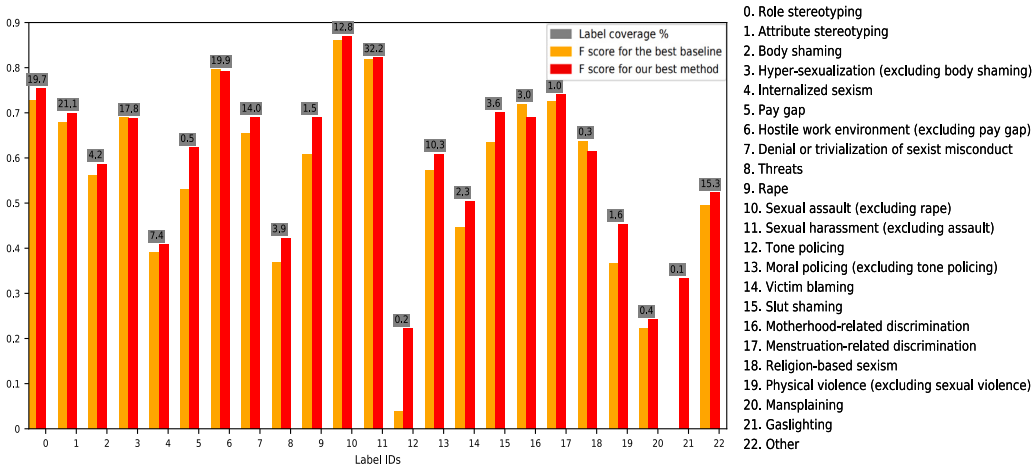

Fig. 7. Class-wise sexism classification F-scores for the best-performing baseline (Parikh et al., 2019) and our best method (*S-det, Sarc-det, E-pred, Topic-p, Cl-pred with EBCEL-cor and WLTL*). Each gray box contains the % of the training data samples that the corresponding label applies to.

Table 6
Performance variation across #labels per post.

#labels per post	Approach	F_{ins}	F_{mac}	F_{mic}	Acc	SA
1	Best proposed method	**0.715**	**0.494**	**0.619**	**0.608**	**0.412**
	Best baseline method	0.667	0.414	0.554	0.537	0.314
2	Best proposed method	**0.752**	**0.592**	**0.714**	**0.621**	**0.286**
	Best baseline method	0.719	0.532	0.667	0.575	0.215
3	Best proposed method	**0.755**	**0.640**	**0.735**	**0.619**	**0.209**
	Best baseline method	0.739	0.535	0.705	0.592	0.174
4	Best proposed method	**0.756**	**0.670**	**0.741**	**0.613**	**0.113**
	Best baseline method	0.721	0.592	0.721	0.594	0.094
5	Best proposed method	**0.756**	0.666	**0.746**	**0.607**	0.061
	Best baseline method	0.740	**0.716**	0.728	0.592	**0.091**
6	Best proposed method	**0.802**	**0.877**	**0.800**	**0.672**	0.156
	Best baseline method	0.800	0.789	0.784	0.671	**0.167**

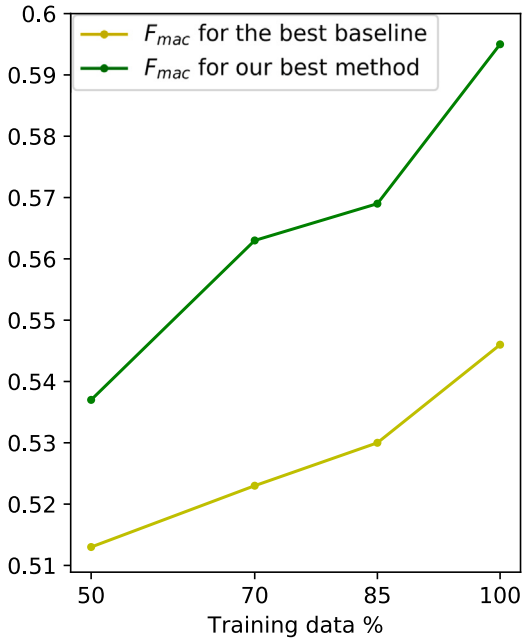

Fig. 8. F_{mac} for varying training data percentages.

Table 7
Accounts of sexism correctly classified by our best multitask method with WLTL approach but not the best baseline.

Account of sexism	True labels	Avg. label coverage
When my 15 year old sister went dress shopping for a disco my mother kindly reminder her to not get something tarty and to be careful of looking like a tart.	Internalized sexism, Moral policing (excluding tone policing), Slut shaming	7.11%
Media labels see us only as 'mums' or 'babes'.	Motherhood-related discrimination, Role stereotyping, Hyper-sexualization (excluding body shaming), Moral policing (excluding tone policing)	12.71%
In our University, a male lecturer speaking to my female office-mate about her teaching responsibilities 'Why aren't you teaching anything on gender then?' Office-mate 'All my stuff incorporates concerns about gender but I teach design history'. Male lecturer 'I thought you would be teaching all that women's stuff'.	Role stereotyping, Attribute stereotyping, Hostile work environment (excluding pay gap)	20.28%
I told her about the app my husband (a writer) and I were producing together with the help of a developer. "Is your husband technically oriented then?" Um, no, actually I'm the more tech-oriented one, and it was my idea.	Role stereotyping, Attribute stereotyping, Internalized sexism	16.10%

Table 7 provides accounts of sexism from the test set for which our combined method made the right predictions but the best baseline did not. For each account, the associated labels and the average of their coverage values are also reported. Our method can be seen to work well across different labels and coverage values.

6. Conclusion

We investigated the 23-class fine-grained classification of accounts of sexism in this work. To address this, we explored a knowledge-based cascaded multi-task learning framework that allows for shared learning across multiple tasks using homogeneous and heterogeneous auxiliary tasks. Homogeneous auxiliary tasks are automatically set up through unsupervised learning from unlabeled accounts of sexism and weak labeling. Heterogeneous tasks are considered to extract more cues from the cross-domain data. The knowledge module is incorporated into the framework in order to fuse the external knowledge (meaning of the domain-specific keywords) representations into a learning process. We also explore the weakly labeled accounts of sexism and perform transfer learning where we train the models sequentially using weakly labeled data and labeled data at different levels of granularity. Moreover, we formulate loss functions through which we seek to utilize correlations existing between labels in the training data.

Our proposed methods outperform various traditional machine learning and deep learning baselines across many standard metrics, including Subset Accuracy, the strictest metric. The best proposed objective function is *EBCEL-cor* which bring together the actual conditional co-occurrence probabilities and the corresponding model-based probability estimates. Our proposed multi-task methods with *EBCEL-cor* shows that the complementary signals will help to perform the better classification. Our best multi-task method is *S-det, Sarc-det, E-pred, Topic-p, Cl-pred with EBCEL-cor* which involves all the homogeneous and heterogeneous auxiliary tasks. The performance of this best method is further enhanced when combined with WLTL approach; it achieves an instance-based F-score (F_{ins}) of 0.740, F_{mac} of 0.595 and SA of 0.309 whereas the best baseline produces an instance-based F-score (F_{ins}) of 0.714, F_{mac} of 0.546 and SA of 0.242.

Future work aims to explore few-shot learning for sexism classification, as we have a few samples for some categories in our fine-grained classification setup. Another possible direction is to explore multi-lingual sexism detection and classification as social media presence in languages other than English. We can investigate the zero-shot learning for this where we can consider training data from all the other languages and test on one target language.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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